ELITE IGCSE ACADEMY

Private Past Paper Solutions

Specimen 4WM2H Worked Solutions

Specimen | 4WM2H | Unit 2

29 questions ▪ question image followed by worked solution

PREPARED BY

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TEACHER EDITION

Worked answers included

PRIVATE

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PAST PAPER QUESTION

Q#: 1 Marks: 3

TOPIC: **Statistics Toolkit**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

1 Here are five cards, where x represents a number.

15	7	–2	23	x

The mean of the five numbers is 12

Work out the value of x

$x =$ ....

(Total for Question 1 is 3 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 3

TOPIC: **Statistics Toolkit**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

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Topic check. Statistics Toolkit. The tag is correct.

Solution

The mean is 12, so the total of the five numbers is

$$5 \times 12 = 60$$

The known numbers add to

$$15 + 7 - 2 + 23 = 43$$

Therefore

$$43 + x = 60$$

$$x = 17$$

FINAL ANSWER

$x = 17$.

PAST PAPER QUESTION

Q#: 2 Marks: 4

TOPIC: **Prime Factors, HCF & LCM**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

2 (a) Find the highest common factor (HCF) of 56 and 84
Show your working clearly.

....

(2)

(b) Find the lowest common multiple (LCM) of 60 and 72
Show your working clearly.

....

(2)

(Total for Question 2 is 4 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 4

TOPIC: **Prime Factors, HCF & LCM**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Prime Factors, HCF & LCM. The tag is correct.

Solution

$$56 = 2^3 \times 7, \qquad 84 = 2^2 \times 3 \times 7$$

The common prime factors are 2^2 and 7, so

$$\text{HCF} = 2^2 \times 7 = 28$$

Also

$$60 = 2^2 \times 3 \times 5, \qquad 72 = 2^3 \times 3^2$$

Use the highest powers of each prime:

$$\text{LCM} = 2^3 \times 3^2 \times 5 = 360$$

FINAL ANSWER

HCF $= 28$, LCM $= 360$.

PAST PAPER QUESTION

Q#: 3 Marks: 2

Topic: **Algebraic Roots & Indices**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

3 $\frac{2^{k}}{4^{n}} = 2^{x}$

Find an expression for x in terms of k and n

$x =$ ....

(Total for Question 3 is 2 marks)

PAST PAPER SOLUTION

Q#: 3 Marks: 2

TOPIC: **Algebraic Roots & Indices**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

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Topic check. Algebraic Roots & Indices. The tag is correct because the question uses index laws with letters.

Solution

$$4^n = (2^2)^n = 2^{2n}$$

So

$$\frac{2^k}{4^n} = \frac{2^k}{2^{2n}} = 2^{k-2n}$$

Since this equals 2^x,

$$x = k - 2n$$

FINAL ANSWER

$x = k - 2n$.

PAST PAPER QUESTION

Q#: 4 Marks: 4

TOPIC: **Angles in Polygons & Parallel Lines**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

4 The diagram shows parts of three regular polygons, **A**, **B** and **C**, meeting at a point.

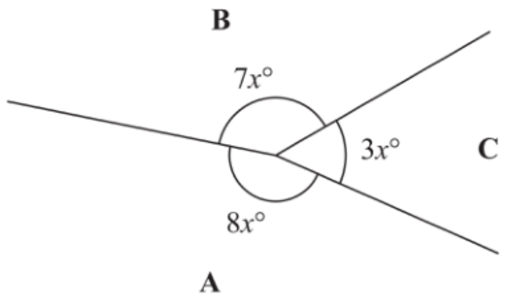

Diagram **NOT** accurately drawn

Polygon **B** has n sides.
Work out the value of n

$n =$ ....

(Total for Question 4 is 4 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 4

TOPIC: **Angles in Polygons & Parallel Lines**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

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Topic check. Angles in Polygons & Parallel Lines. The main idea is angles around a point and the interior angle of a regular polygon.

Solution

Angles around a point add to $360°$:

$$8x + 7x + 3x = 360$$

$$18x = 360$$

$$x = 20$$

The interior angle of polygon B is

$$7x = 7(20) = 140°$$

For a regular n-sided polygon,

$$\frac{180(n-2)}{n} = 140$$

$$180n - 360 = 140n$$

$$40n = 360$$

$$n = 9$$

FINAL ANSWER

$n = 9$.

PAST PAPER QUESTION

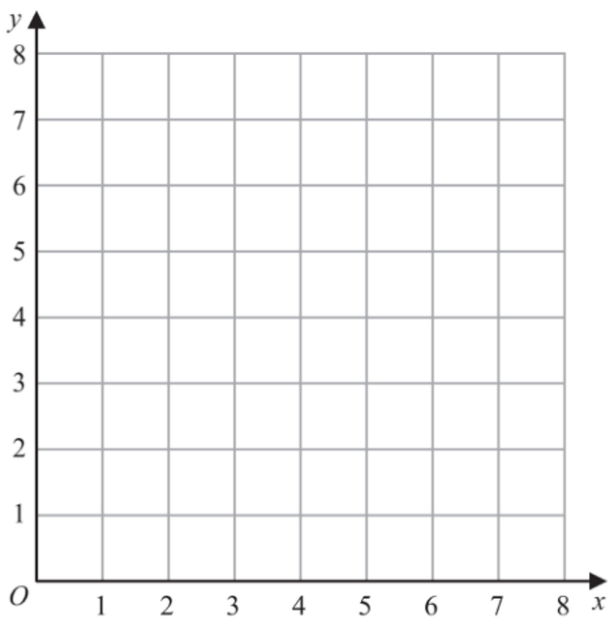
Q#: 5 Marks: 4

TOPIC: **Graphing Inequalities**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

5 (a) On the grid, draw and label with its equation the straight line with equation

(i) $y = 1$ (ii) $x = 2$ (iii) $x + y = 7$



(3)

(b) Show, by shading on the grid, the region that satisfies **all three** of the inequalities

$y \geqslant 1$ $x \geqslant 2$ $x + y \leqslant 7$

Label the region **R**

(1)

(Total for Question 5 is 4 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 4

TOPIC: **Graphing Inequalities**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

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Topic check. Graphing Inequalities. The tag is correct.

Solution

Draw the three boundary lines:

$$y = 1, \qquad x = 2, \qquad x + y = 7$$

For $x + y = 7$, use points such as $(0, 7)$ and $(7, 0)$.
The required region is:

$$y \geq 1, \qquad x \geq 2, \qquad x + y \leq 7$$

So shade the triangular region above $y = 1$, to the right of $x = 2$, and below the line $x + y = 7$. Its vertices are

$$(2, 1), \quad (2, 5), \quad (6, 1)$$

FINAL ANSWER

the shaded region R is the triangle with vertices $(2, 1)$, $(2, 5)$, and $(6, 1)$.

PAST PAPER QUESTION Q#: 6 Marks: 3

TOPIC: **Statistics Toolkit**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

6 Here are some integers where $a < b < c < d$

$$a \quad b \quad c \quad d \quad d \quad d$$

The mode of the integers is 9
The range of the integers is 4
The median of the integers is 8

Work out the value of a, the value of b, the value of c and the value of d

$a =$ ....

$b =$ ....

$c =$ ....

$d =$ ....

(Total for Question 6 is 3 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 3

TOPIC: **Statistics Toolkit**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

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Topic check. Statistics Toolkit. The tag is correct.

Solution

The mode is 9, and d appears three times, so

$$d = 9$$

The range is 4, so

$$d - a = 4$$

$$9 - a = 4$$

$$a = 5$$

There are six values, so the median is the mean of the 3rd and 4th values:

$$\frac{c + d}{2} = 8$$

$$\frac{c + 9}{2} = 8$$

$$c = 7$$

Since $a < b < c < d$, the remaining integer is $b = 6$.

FINAL ANSWER

$a = 5,\ b = 6,\ c = 7,\ d = 9.$

PAST PAPER QUESTION

Q#: 7 Marks: 4

TOPIC: **Standard & Compound Units**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

7 A cylinder is placed on the ground.

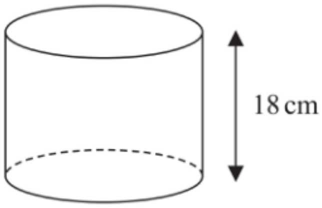

Diagram **NOT** accurately drawn

The height of the cylinder is 18 cm

The force exerted by the cylinder on the ground is 72 newtons.
The pressure on the ground due to the cylinder is 1.4 newtons/cm^2

$$\boxed{\text{pressure} = \frac{\text{force}}{\text{area}}}$$

Work out the volume of the cylinder.
Give your answer correct to 3 significant figures.

.... cm^3

(Total for Question 7 is 4 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 4

TOPIC: **Standard & Compound Units**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

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Topic check. Standard & Compound Units. The pressure formula is the key signal, with volume used at the end.

Solution

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

So the area of the base is

$$\text{area} = \frac{72}{1.4} = 51.428571\ldots$$

The volume is

$$51.428571\ldots \times 18 = 925.714285\ldots$$

Correct to 3 significant figures,

$$925.714\ldots \approx 926$$

FINAL ANSWER

926 cm^3.

PAST PAPER QUESTION

Q#: 8 Marks: 6

TOPIC: **Compound Interest & Depreciation**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

8 In 2021, the value of Asha's apartment was 634 400 euros.
The value of Asha's apartment had increased by 4% from its value in 2020

(a) Work out the value of Asha's apartment in 2020

.... euros
(3)

Pam bought a boat.

In each year after Pam bought the boat, the value of the boat depreciated by 15%

(b) Work out the total percentage by which the value of the boat had depreciated by the end of the second year after Pam bought the boat.

.... %
(3)

(Total for Question 8 is 6 marks)

PAST PAPER SOLUTION

Q#: 8 Marks: 6

TOPIC: **Compound Interest & Depreciation**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

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Topic check. Compound Interest & Depreciation. The tag is correct.

Solution

For part (a), the 2021 value is 104% of the 2020 value.

$$2020 \text{ value} = \frac{634400}{1.04} = 610000$$

For part (b), after one year the multiplier is 0.85. After two years:

$$0.85^2 = 0.7225$$

So 72.25% of the value remains, and the total depreciation is

$$100\% - 72.25\% = 27.75\%$$

FINAL ANSWER

610000 euros; 27.75%.

PAST PAPER QUESTION

Q#: 9 Marks: 2

TOPIC: **Powers, Roots & Standard Form**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

9 (a) Write 0.000089 in standard form.

....

(1)

(b) Write 8.34×10^4 as an ordinary number.

....

(1)

(Total for Question 9 is 2 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 2

TOPIC: **Powers, Roots & Standard Form**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Powers, Roots & Standard Form. The tag is correct.

Solution

$$0.000089 = 8.9 \times 10^{-5}$$

Also

$$8.34 \times 10^{4} = 83400$$

FINAL ANSWER

8.9×10^{-5}, 83400.

PAST PAPER QUESTION

Q#: 10 Marks: 5

TOPIC: **Ratio Problem Solving**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

10 Payel makes 300 celebration cards so that

number of birthday cards : number of anniversary cards : number of congratulations cards = 7 : 5 : 3

$\frac{2}{5}$ of the birthday cards have numbers on them.

36% of the anniversary cards have numbers on them.
None of the congratulations cards have numbers on them.

Work out what fraction of the 300 cards have numbers on them.
Give your answer in its simplest form.

....

(Total for Question 10 is 5 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 5

TOPIC: **Ratio Problem Solving**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

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Topic check. Ratio Problem Solving. The ratio is needed before the fraction and percentage work.

Solution

The total number of parts is

$$7 + 5 + 3 = 15$$

Each part is

$$300 \div 15 = 20$$

So the numbers of cards are:

$$140, \quad 100, \quad 60$$

The birthday cards with numbers are

$$\frac{2}{5} \times 140 = 56$$

The anniversary cards with numbers are

$$36\% \times 100 = 36$$

So the total with numbers is

$$56 + 36 = 92$$

The fraction of all cards is

$$\frac{92}{300} = \frac{23}{75}$$

FINAL ANSWER

$\frac{23}{75}$.

PAST PAPER QUESTION

Q#: 11 Marks: 3

TOPIC: **Simultaneous Equations**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

11 Solve the simultaneous equations

$$7x + 3y = 3$$
$$3x - y = 7$$

Show clear algebraic working.

$x =$ ....

$y =$ ....

(Total for Question 11 is 3 marks)

PAST PAPER SOLUTION

Q#: 11 Marks: 3

TOPIC: **Simultaneous Equations**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Simultaneous Equations. The tag is correct.

Solution

From

$$3x - y = 7$$

$$y = 3x - 7$$

Substitute into $7x + 3y = 3$:

$$7x + 3(3x - 7) = 3$$

$$16x - 21 = 3$$

$$16x = 24$$

$$x = 1.5$$

Then

$$y = 3(1.5) - 7 = -2.5$$

FINAL ANSWER

$x = 1.5,\ y = -2.5.$

PAST PAPER QUESTION

Q#: 12 Marks: 5

TOPIC: **Percentages**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

12 Zimo is going on holiday.
He makes 3 separate payments to cover the total cost of his holiday.

The following table shows how much money Zimo pays to the holiday company.

Payment	**Amount paid**
Payment 1	$\frac{2}{5}$ of the total cost
Payment 2	45% of the total cost
Payment 3	$405

Work out how much Zimo has to pay for Payment 2

$..

(Total for Question 12 is 5 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 5

Topic: **Percentages**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Percentages. The tag is correct.

Solution

$$\frac{2}{5} = 40\%$$

So Payments 1 and 2 together are

$$40\% + 45\% = 85\%$$

This means Payment 3 is

$$100\% - 85\% = 15\%$$

So

$$15\% = 405$$

$$1\% = 27$$

Payment 2 is 45%:

$$45 \times 27 = 1215$$

FINAL ANSWER

\$1215.

PAST PAPER QUESTION Q#: 13 Marks: 4

TOPIC: **Functions**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

13 The function f is defined as

$$f : x \mapsto \frac{2x}{x-6}$$

(a) Find f(10)

....

(1)

(b) Express the inverse function f^{-1} in the form $f^{-1} : x \mapsto \ldots$

$f^{-1} : x \mapsto$ ....

(3)

PAST PAPER SOLUTION

Q#: 13 Marks: 4

TOPIC: **Functions**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

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Topic check. Functions. The tag is correct.

Solution

$$f(10) = \frac{2(10)}{10-6} = \frac{20}{4} = 5$$

For the inverse, let

$$y = \frac{2x}{x-6}$$

$$y(x-6) = 2x$$

$$xy - 6y = 2x$$

$$x(y-2) = 6y$$

$$x = \frac{6y}{y-2}$$

So

$$f^{-1} : x \mapsto \frac{6x}{x-2}$$

FINAL ANSWER

5, and $f^{-1} : x \mapsto \frac{6x}{x-2}$.

PAST PAPER QUESTION

Q#: 14 **Marks: 5**

TOPIC: **Circle Theorems**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

14 *A*, *B*, C and *D* are points on a circle, centre *O*

EBF is the tangent to the circle at *B*

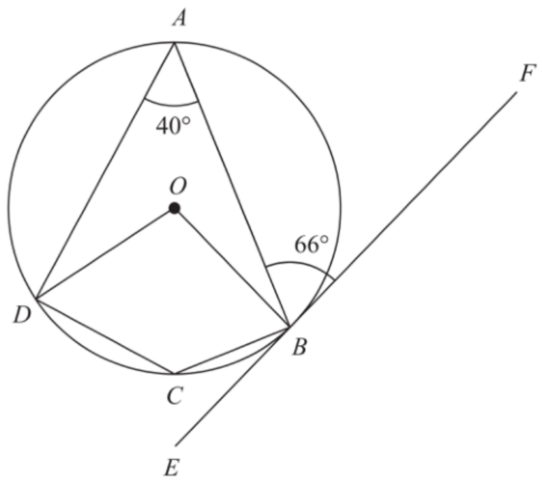


Diagram **NOT** accurately drawn

(a) (i) Work out the size of angle *DCB*

....

(1)

(ii) Give a reason for your answer to (a)(i)

....

....

(1)

(b) Work out the size of angle *ADO*

..............................°

(3)

(Total for Question 14 is 5 marks)

PAST PAPER SOLUTION

Q#: 14 Marks: 5

TOPIC: **Circle Theorems**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Circle Theorems. The tag is correct.

Solution

In cyclic quadrilateral $ADCB$, opposite angles add to 180°.

$$\angle DCB = 180^\circ - 40^\circ = 140^\circ$$

Reason: opposite angles in a cyclic quadrilateral add to 180°.
By the alternate segment theorem,

$$\angle ADB = 66^\circ$$

In triangle ADB,

$$\angle ABD = 180^\circ - 40^\circ - 66^\circ = 74^\circ$$

The angle at the centre is twice the angle at the circumference standing on the same chord AD:

$$\angle AOD = 2(74^\circ) = 148^\circ$$

Since $OA = OD$, triangle AOD is isosceles:

$$\angle ADO = \frac{180^\circ - 148^\circ}{2} = 16^\circ$$

FINAL ANSWER
$\angle DCB = 140^\circ$; reason: opposite angles in a cyclic quadrilateral; $\angle ADO = 16^\circ$.

PAST PAPER QUESTION

Q#: 15 Marks: 7

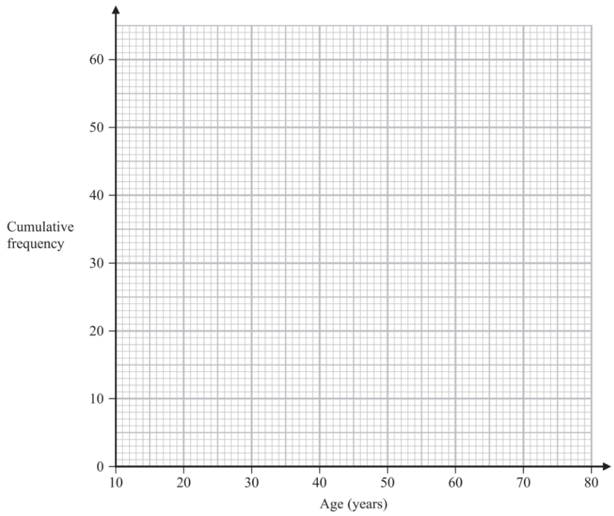
TOPIC: **Cumulative Frequency Diagrams**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

15 The cumulative frequency table shows information about the ages of 60 people who went to a gym on Saturday.

Age (a years)	Cumulative frequency
$10 < a \leqslant 20$	13
$10 < a \leqslant 30$	36
$10 < a \leqslant 40$	42
$10 < a \leqslant 50$	47
$10 < a \leqslant 60$	52
$10 < a \leqslant 70$	56
$10 < a \leqslant 80$	60

(a) On the grid, draw a cumulative frequency graph for the information in the table.



(2)

(b) Use your graph to find an estimate for the median of the ages of these people.

.... years

(1)

(c) Use your graph to find an estimate for the interquartile range of the ages of these people.

.... years

(2)

(d) Use your graph to find an estimate for the number of these people who are older than 55 years.

....

(2)

(Total for Question 15 is 7 marks)

PAST PAPER SOLUTION

Q#: 15 Marks: 7

TOPIC: **Cumulative Frequency Diagrams**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Cumulative Frequency Diagrams. The tag is correct.

Solution

Plot the cumulative frequency points and join with a smooth increasing curve:

$$(10, 0),\ (20, 13),\ (30, 36),\ (40, 42),\ (50, 47),\ (60, 52),\ (70, 56),\ (80, 60)$$

There are 60 people, so the median is read at cumulative frequency 30.
Using linear interpolation between $(20, 13)$ and $(30, 36)$:

$$20 + \frac{30 - 13}{36 - 13} \times 10 \approx 27.4$$

So the median is about 27 years.
For the interquartile range, read Q_1 at cumulative frequency 15 and Q_3 at cumulative frequency 45:

$$Q_1 \approx 20.9, \qquad Q_3 \approx 46$$

$$\text{IQR} \approx 46 - 20.9 = 25.1$$

So the IQR is about 25 years.
At age 55, the cumulative frequency is about 49.5, so the number older than 55 is

$$60 - 49.5 \approx 10.5$$

This gives about 11 people.

FINAL ANSWER
median about 27 years, IQR about 25 years, about 11 people older than 55.

PAST PAPER QUESTION

Q#: 16 Marks: 4

TOPIC: **Direct & Inverse Proportion**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

16 M is directly proportional to h^3
$M = 4$ when $h = 0.5$

Find the value of h when $M = 500$

$h =$ ....

(Total for Question 16 is 4 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 4

TOPIC: **Direct & Inverse Proportion**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Direct & Inverse Proportion. The tag is correct.

Solution

Since M is directly proportional to h^3,

$$M = kh^3$$

Use $M = 4$ when $h = 0.5$:

$$4 = k(0.5)^3$$

$$4 = 0.125k$$

$$k = 32$$

Now use $M = 500$:

$$500 = 32h^3$$

$$h^3 = 15.625$$

$$h = 2.5$$

FINAL ANSWER

$h = 2.5$.

PAST PAPER QUESTION

Q#: 17 Marks: 5

TOPIC: **Differentiation**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

17 A particle P moves along a straight line.
The fixed point O lies on this line.

The displacement of P from O at time t seconds, $t \geqslant 1$, is s metres where

$$s = 4t^2 + \frac{125}{t}$$

The velocity of P at time t seconds, $t \geqslant 1$, is v m/s

Work out the distance of P from O at the instant when $v = 0$

.... m

(Total for Question 17 is 5 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 5

TOPIC: **Differentiation**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Differentiation. The tag is correct.

Solution

$$s = 4t^2 + \frac{125}{t}$$

Differentiate to find velocity:

$$v = \frac{ds}{dt} = 8t - \frac{125}{t^2}$$

Set $v = 0$:

$$8t - \frac{125}{t^2} = 0$$

$$8t^3 = 125$$

$$t^3 = \frac{125}{8}$$

$$t = 2.5$$

Now find s:

$$s = 4(2.5)^2 + \frac{125}{2.5}$$

$$s = 25 + 50 = 75$$

FINAL ANSWER

75 m.

PAST PAPER QUESTION

Q#: 18 Marks: 3

TOPIC: **Solving Inequalities**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

18 Solve the inequality $2y^2 - 7y - 30 \leqslant 0$
Show your working clearly.

....

(Total for Question 18 is 3 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 3

TOPIC: **Solving Inequalities**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

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Topic check. Solving Inequalities. The tag is correct.

Solution

Factorise:

$$2y^2 - 7y - 30 = (2y + 5)(y - 6)$$

So

$$(2y + 5)(y - 6) \leq 0$$

The roots are

$$y = -\frac{5}{2}, \qquad y = 6$$

The quadratic opens upwards, so it is less than or equal to zero between the roots.

FINAL ANSWER

$-\frac{5}{2} \leq y \leq 6$.

PAST PAPER QUESTION Q#: 19 Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

19 The diagram shows two similar metal statues.

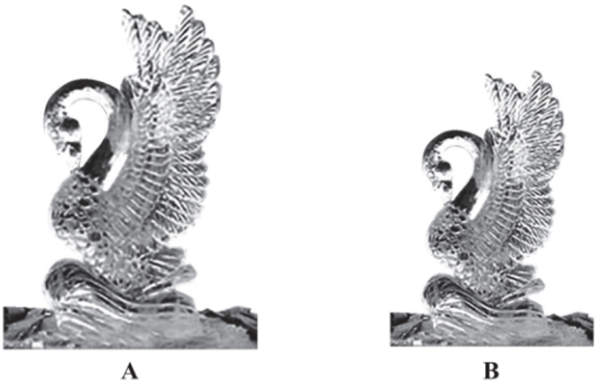

Diagram **NOT** accurately drawn

The volume of statue **B** is 20% less than the volume of statue **A**
The surface area of statue **B** is k% less than the surface area of statue **A**

Work out the value of k
Give your answer correct to 3 significant figures.

k = ...

(Total for Question 19 is 4 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Area & Volume of Similar Shapes. The tag is correct.

Solution

The volume of statue B is 80% of the volume of statue A, so the volume scale factor is

$$0.8$$

For similar solids:

$$\text{linear scale factor} = \sqrt[3]{0.8}$$

The surface area scale factor is therefore

$$\left(\sqrt[3]{0.8}\right)^2 = 0.8^{2/3}$$

$$0.8^{2/3} = 0.861773\ldots$$

So the percentage decrease in surface area is

$$(1 - 0.861773\ldots) \times 100 = 13.8226\ldots$$

Correct to 3 significant figures,

$$k = 13.8$$

FINAL ANSWER

$k = 13.8$.

PAST PAPER QUESTION

Q#: 20 Marks: 5

TOPIC: **Simultaneous Equations**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

20 Solve the simultaneous equations

$$x - 2y = 3$$
$$x^2 - y^2 + 2x = 10$$

Show clear algebraic working.

....

(Total for Question 20 is 5 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 5

TOPIC: **Simultaneous Equations**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

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Topic check. Simultaneous Equations. The tag is correct; this is a linear and quadratic simultaneous equation.

Solution

From

$$x - 2y = 3$$

$$x = 2y + 3$$

Substitute into $x^2 - y^2 + 2x = 10$:

$$(2y+3)^2 - y^2 + 2(2y+3) = 10$$

$$4y^2 + 12y + 9 - y^2 + 4y + 6 = 10$$

$$3y^2 + 16y + 5 = 0$$

$$(3y+1)(y+5) = 0$$

So

$$y = -\frac{1}{3} \quad \text{or} \quad y = -5$$

If $y = -\frac{1}{3}$,

$$x = 2\left(-\frac{1}{3}\right) + 3 = \frac{7}{3}$$

If $y = -5$,

$$x = 2(-5) + 3 = -7$$

FINAL ANSWER

$\left(\frac{7}{3}, -\frac{1}{3}\right)$ and $(-7, -5)$.

PAST PAPER QUESTION

Q#: 20 Marks: 5

TOPIC: **Simultaneous Equations**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

20 Solve the simultaneous equations

$$x - 2y = 3$$
$$x^2 - y^2 + 2x = 10$$

Show clear algebraic working.

....

(Total for Question 20 is 5 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 5

TOPIC: **Simultaneous Equations**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Simultaneous Equations. The tag is correct; this is a linear and quadratic simultaneous equation.

Solution

From

$$x - 2y = 3$$

$$x = 2y + 3$$

Substitute into $x^2 - y^2 + 2x = 10$:

$$(2y+3)^2 - y^2 + 2(2y+3) = 10$$

$$4y^2 + 12y + 9 - y^2 + 4y + 6 = 10$$

$$3y^2 + 16y + 5 = 0$$

$$(3y+1)(y+5) = 0$$

So

$$y = -\frac{1}{3} \quad \text{or} \quad y = -5$$

If $y = -\frac{1}{3}$,

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FINAL ANSWER

$\left(\frac{7}{3}, -\frac{1}{3}\right)$ and $(-7, -5)$.

PAST PAPER QUESTION Q#: 21 Marks: 3

TOPIC: **Algebraic Fractions**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

21 $a = \dfrac{14}{3x - 7}$ $\qquad x = \dfrac{7}{4y - 3}$

Express a in the form $\dfrac{py + q}{ry + s}$ where p, q, r and s are integers.

Give your answer in its simplest form.

$a =$ ....

(Total for Question 21 is 3 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 3

TOPIC: **Algebraic Fractions**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Algebraic Fractions. The tag is correct because the task is substitution and simplification of algebraic fractions.

Solution

$$a = \frac{14}{3x-7}, \qquad x = \frac{7}{4y-3}$$

Substitute for x:

$$a = \frac{14}{3\left(\frac{7}{4y-3}\right) - 7}$$

Simplify the denominator:

$$3\left(\frac{7}{4y-3}\right) - 7 = \frac{21}{4y-3} - 7$$

$$= \frac{21 - 7(4y-3)}{4y-3} = \frac{42-28y}{4y-3}$$

So

$$a = \frac{14}{\frac{42-28y}{4y-3}} = \frac{14(4y-3)}{42-28y}$$

$$a = \frac{4y-3}{3-2y}$$

FINAL ANSWER

$a = \dfrac{4y-3}{3-2y}$.

PAST PAPER QUESTION

Q#: 21 Marks: 3

Topic: **Algebraic Fractions**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

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Express a in the form $\dfrac{py+q}{ry+s}$ where p, q, r and s are integers.

Give your answer in its simplest form.

$a =$ ....

(Total for Question 21 is 3 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 3

Topic: **Algebraic Fractions**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

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$$= \frac{21 - 7(4y - 3)}{4y - 3} = \frac{42 - 28y}{4y - 3}$$

So

$$a = \frac{14}{\frac{42-28y}{4y-3}} = \frac{14(4y - 3)}{42 - 28y}$$

$$a = \frac{4y - 3}{3 - 2y}$$

FINAL ANSWER

$a = \dfrac{4y - 3}{3 - 2y}$.

PAST PAPER QUESTION Q#: 22 Marks: 5

Topic: **Volume & Surface Area**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

22 A solid is made from a cone and a hemisphere.

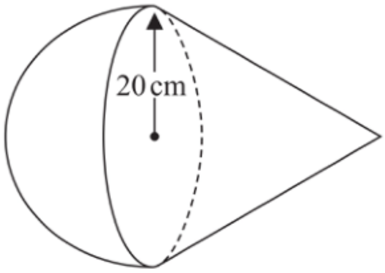

Diagram **NOT** accurately drawn

The circular plane face of the hemisphere coincides with the circular base of the cone.
The radius of the hemisphere and the radius of the circular base of the cone are both 20 cm

The curved surface area of the cone is $580\pi\,\text{cm}^2$

The volume of the solid is $k\pi\,\text{cm}^3$

Work out the exact value of k

$k =$ ....

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 5

TOPIC: **Volume & Surface Area**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Volume & Surface Area. The tag is correct.

Solution

The curved surface area of a cone is

$$\pi r l$$

Here $r = 20$ and $\pi r l = 580\pi$, so

$$20l = 580$$

$$l = 29$$

Use Pythagoras to find the height of the cone:

$$h^2 = 29^2 - 20^2$$

$$h^2 = 841 - 400 = 441$$

$$h = 21$$

Volume of the cone:

$$\frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(20)^2(21) = 2800\pi$$

Volume of the hemisphere:

$$\frac{2}{3}\pi r^3 = \frac{2}{3}\pi(20)^3 = \frac{16000}{3}\pi$$

Total volume:

$$2800\pi + \frac{16000}{3}\pi = \frac{24400}{3}\pi$$

Since the volume is $k\pi$,

$$k = \frac{24400}{3}$$

FINAL ANSWER

$k = \dfrac{24400}{3}$.

PAST PAPER QUESTION

TOPIC: **Volume & Surface Area**

Q#: 22 Marks: 5

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

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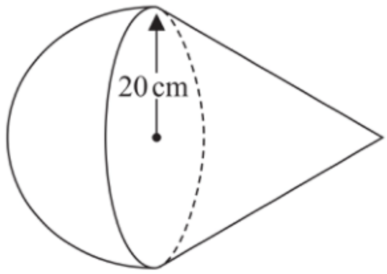

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Work out the exact value of k

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(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 5

TOPIC: **Volume & Surface Area**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

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FINAL ANSWER

$k = \dfrac{24400}{3}$.

PAST PAPER QUESTION

Q#: 23 Marks: 5

TOPIC: **Vectors**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

23

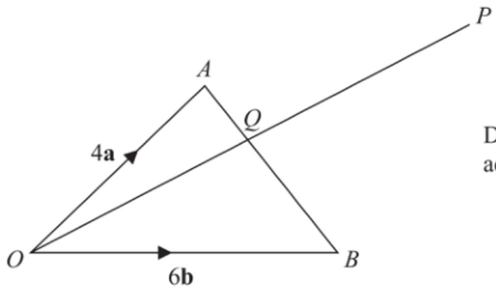


Diagram **NOT** accurately drawn

OAB is a triangle.
Q is the point on *AB* such that *OQP* is a straight line.

$\overrightarrow{OA} = 4\mathbf{a}$ $\quad\overrightarrow{OB} = 6\mathbf{b}$ $\quad\overrightarrow{AP} = 2\mathbf{a} + 8\mathbf{b}$

Using a vector method, find the ratio $AQ : QB$

$AQ : QB =$ ....

(Total for Question 23 is 5 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 5

TOPIC: **Vectors**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Vectors. The tag is correct.

Solution

$$\overrightarrow{OA} = 4a, \qquad \overrightarrow{OB} = 6b, \qquad \overrightarrow{AP} = 2a + 8b$$

So

$$\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP}$$

$$\overrightarrow{OP} = 4a + (2a + 8b) = 6a + 8b$$

Since O, Q, P are collinear, let

$$\overrightarrow{OQ} = \lambda(6a + 8b)$$

Also Q lies on AB. Let

$$\overrightarrow{AQ} = \mu\overrightarrow{AB}$$

Then

$$\overrightarrow{AB} = 6b - 4a$$

$$\overrightarrow{OQ} = 4a + \mu(6b - 4a)$$

$$\overrightarrow{OQ} = (4 - 4\mu)a + 6\mu b$$

Compare coefficients with

$$\lambda(6a + 8b) = 6\lambda a + 8\lambda b$$

So

$$4 - 4\mu = 6\lambda, \qquad 6\mu = 8\lambda$$

From $6\mu = 8\lambda$,

$$\lambda = \frac{3}{4}\mu$$

Substitute:

$$4 - 4\mu = 6\left(\frac{3}{4}\mu\right)$$

$$4 - 4\mu = \frac{9}{2}\mu$$

$$8 = 17\mu$$

$$\mu = \frac{8}{17}$$

Therefore

$$AQ : QB = \frac{8}{17} : \frac{9}{17} = 8 : 9$$

FINAL ANSWER
$AQ : QB = 8 : 9.$

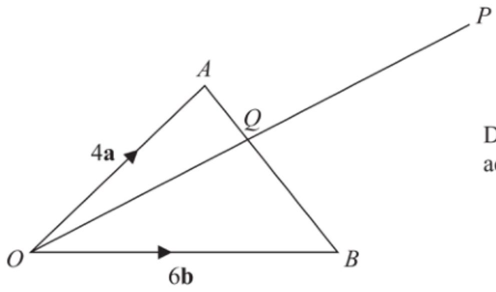
PAST PAPER QUESTION

Q#: 23 Marks: 5

TOPIC: **Vectors**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

23



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$AQ : QB =$ ....

(Total for Question 23 is 5 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 5

TOPIC: **Vectors**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Vectors. The tag is correct.

Solution

$$\overrightarrow{OA} = 4a, \qquad \overrightarrow{OB} = 6b, \qquad \overrightarrow{AP} = 2a + 8b$$

So

$$\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP}$$

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Substitute:

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$$4 - 4\mu = \frac{9}{2}\mu$$

$$8 = 17\mu$$

$$\mu = \frac{8}{17}$$

Therefore

$$AQ : QB = \frac{8}{17} : \frac{9}{17} = 8 : 9$$

FINAL ANSWER
$AQ : QB = 8 : 9.$

PAST PAPER QUESTION

Q#: 24 Marks: 5

TOPIC: **Sequences**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

24 The sum of the first 10 terms of an arithmetic series is 4 times the sum of the first 5 terms of the same series.
The 8th term of this series is 45

Find the first term of this series.
Show clear algebraic working.

....

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 5

TOPIC: **Sequences**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Sequences. The tag is correct.

Solution

Let the first term be a and the common difference be d.

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

So

$$S_{10} = 5(2a + 9d)$$

and

$$S_5 = \frac{5}{2}(2a + 4d)$$

Given $S_{10} = 4S_5$:

$$5(2a + 9d) = 4\left(\frac{5}{2}(2a + 4d)\right)$$

$$10a + 45d = 20a + 40d$$

$$5d = 10a$$

$$d = 2a$$

The 8th term is 45:

$$a + 7d = 45$$

Substitute $d = 2a$:

$$a + 14a = 45$$

$$15a = 45$$

$$a = 3$$

FINAL ANSWER

the first term is 3.

PAST PAPER QUESTION

Q#: 24 Marks: 5

TOPIC: **Sequences**

Specimen 4WM2H - Specimen | 4WM2H | Unit 2

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PAST PAPER SOLUTION

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Substitute $d = 2a$:

$$a + 14a = 45$$

$$15a = 45$$

$$a = 3$$

FINAL ANSWER

the first term is 3.